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Research Article

Geocoding Recommender: An Algorithm to Recommend Optimal Online Geocoding Services for Applications

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Abstract

Today, many services that can geocode addresses are available to domain scientists and researchers, software developers, and end-users. For a number of reasons, including quality of reference database and interpolation technique, a given address geocoded by different services does not often result in the same location. Considering that there are many widely available and accessible geocoding services and that each geocoding service may utilize a different reference database and interpolation technique, selecting a suitable geocoding service that meets the requirements of any application or user is a challenging task. This is especially true for online geocoding services which are often used as black boxes and do not provide knowledge about the reference databases and the interpolation techniques they employ. In this article, we present a geocoding recommender algorithm that can recommend optimal online geocoding services by realizing the characteristics (positional accuracy and match rate) of the services and preferences of the user and/or their application. The algorithm is simulated and analyzed using six popular online geocoding services for different address types (agricultural, commercial, industrial, residential) and preferences (match rate, positional accuracy).

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1 Introduction

Geocoding is one of the basic geospatial operations that converts addresses to geographic locations (e.g. latitude and longitude). Obtaining accurate geocoded locations is important for many application areas such as epidemiology (McElroy et al. 2003; Whitsel et al. 2004; Strickland et al. 2007), environmental science, emergency management, marketing, planning, and location-based services (OGC 2006). Domain scientists, researchers, software developers, and end-users rely on spatial data resulting from geocoding which are the basis for subsequent spatial analysis and mapping. Errors associated with geocoded data are likely to propagate through subsequent research, analysis, modeling, and decision making (Karimi et al. 2004; Zandbergen 2009). Since geocoding errors cannot be avoided completely (Zhang and Goodchild 2002; Karimi et al. 2004), understanding and managing geocoding uncertainties can help users make more realistic estimation (Goldberg et al. 2010).

There are many online geocoding services and offline/conventional geocoding tools available which differ in their reference databases, geocoding algorithms, address parsing strategies, and error reporting methods, resulting in different geocoding qualities (Karimi et al. 2004; Goldberg et al. 2007). Online geocoding services are advantageous over offline/conventional geocoding tools (see Figure 1) in that, for geocoding addresses in online geocoding services, users do not need to prepare reference databases and complex procedures, know and understand the contents of reference databases and the algorithms, and choose suitable services from a large number of options. For example, with *n* conventional geocoding tools having *k* different reference databases and *x* different algorithms, there are n.k.x options, whereas, the number of options to choose an online geocoding service from *n* such services is *n*.

The disadvantages of online geocoding include: less or no control or choice over reference database and parameters like match score, unknown or less trustworthy quality of the geocoded result, and lack of data confidentiality. However, despite the disadvantages, if online geocoding services are preferred over other geocoding approaches, a recommendation method is necessary to choose the optimum service(s) for a given application.

Nevertheless, choosing an optimum online geocoding service is a challenging task for the following reasons (Zandbergen 2009; Roongpiboonsopit and Karimi, 2010a): (1) online geocoding services are often used as black boxes where information about their reference databases, their techniques, and quality of their geocoded results is not available; (2) not every service can adequately satisfy the geocoding preferences of applica-

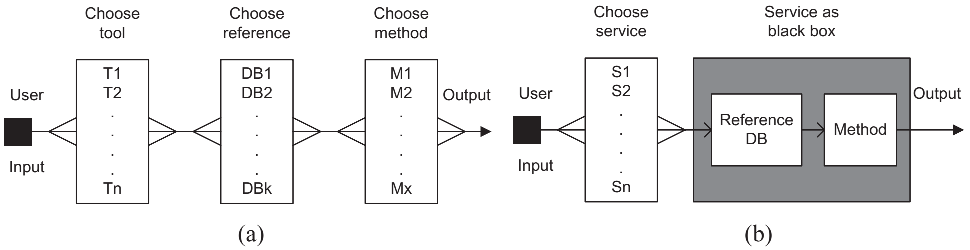

Figure 1 Geocoding options: (a) Conventional geocoding (*n* tools, *k* reference databases and *x* algorithms for each tool); and (b) Online geocoding (*n* services)

tions; (3) no single service can provide similar quality of results for addresses in different geographic areas; and (4) each service has unique capabilities and constraints such as coverage (global vs. local), processing type (single address processing vs. batch address processing), availability of metadata, and number of allowable geocoding requests per day. Therefore, in choosing a suitable service it is imperative to understand the quality of the geocoded results produced by each service and whether or not they conform to the preferences of the underlying application.

Sources of errors that affect quality of geocoding results have been discussed and analyzed extensively in the literature (e.g. Karimi et al. 2004, Yang et al. 2004, Zandbergen and Green 2007). One source of error is the reference database which may contain incomplete, incorrect, inconsistent, and outdated data. Another source of errors is interpolation, which may be based on inappropriate assumptions. In general, quality of geocoding results can be measured by two common metrics: *match rate* and *positional accuracy* (Zandbergen 2009). In our previous work (Roongpiboonsopit and Karimi 2010a), we analyzed and evaluated the quality of geocoding results by various online geocoding services in terms of match rate, positional accuracy, and similarity in different environment settings and for different address types. In another work (Roongpiboonsopit and Karimi 2010b), we conducted a comparative analysis of the quality of geocoding results between two geocoding approaches, *street geocoding* and *rooftop geocoding* (also known as *address point geocoding*). In short, in our previous works we mined online geocoding services to understand their behavior patterns in geocoding addresses with different characteristics. The mining results of online geocoding services, by means of features, parameters, statistics, and behaviors, serve as the basis of the work presented in this article.

In this work, we present and discuss an algorithm that can recommend optimal online geocoding services for applications by using features, parameters, statistics, and behaviors of selected online geocoding services. Previous geocoding studies (e.g. Yang et al. 2004; Zandbergen 2007; Swift et al. 2008; Roongpiboonsopit and Karimi 2010a) focused on comparative analysis of different geocoding services and tools from different perspectives. Currently, however, there is a void in the literature discussing a methodology or service for recommending optimal online geocoding services for given addresses. Furthermore, previous studies did not take an application's geocoding preferences into consideration while analyzing optimal services. It is impractical for a user application to search all available services, understand all their features and capabilities, compare their quality of results, and check requirement matching for each geocoding project. To that end, there is a need for a recommender service that can recommend geocoding services that meet the requirements of the underlying application optimally.

The contribution of this article is an algorithm for recommending optimal online geocoding services that meet an application's requirements. The remainder of the article is organized as follows. Section 2 provides a brief background of online geocoding services. Section 3 presents and describes the recommender algorithm. Simulation of the algorithm and discussion of the results are provided in Section 4. A summary and future research are discussed in Section 5.

2 Background

Currently, there are many online geocoding services available, both free and commercial, with different features and facilities. With online geocoding services, users simply submit a request, usually an address but it could be a point of interest (POI), intersection, ZIP code, city, or county, and receive an output containing geographic coordinates (e.g. latitude and longitude). The output could also be accompanied with level of geocoding accuracy, a full address of the requested input, and other related information. Using online geocoding services does not require the user to choose, maintain, update, or modify the underlying algorithm or reference database. Service providers decide on resources and are responsible for development of services and maintenance of reference databases. Users are mostly concerned about the quality of the geocoded results they receive. Many of the current online geocoding services use street geocoding, some use rooftop geocoding, and some use both (Tetrad 2011, Google Maps 2008). The basic mechanisms, differences, and pros and cons of street geocoding and rooftop geocoding are discussed below.

2.1 Street Geocoding versus Rooftop Geocoding

Street geocoding is an interpolation technique that uses a street network as the reference database (Zandbergen 2007). A street network represents real-world roads as line segments which contain several attributes necessary for geocoding, including address ranges (house numbers or block numbers) on each side of the street, street prefix/suffix, street names, ZIP code, city, state, and country. The attributes of the reference street networks may be different in different countries (Shen 2008). For address types in the U.S. and Canada, street geocoding is accomplished by the following steps: (1) matching the street name; (2) matching the segment that contains the house number; (3) interpolating a point for the address on the segment; and (4) calculating the coordinates of the point (i.e. the estimated location of the address) (Zandbergen 2007). To differentiate the interpolated point on different sides of the segment and at the intersection, side and end off-sets are used, respectively (Zandbergen 2009). Rooftop geocoding is an address-point geocoding technique that employs a point data set as reference database. Each reference point is associated with a physical building or property parcel (usually centroid) of the assigned address.

Rooftop geocoding simply matches the parsed address in the database and returns the associated point in the case of successful match. Since reference databases for rooftop geocoding are not available for many regions, very few online geocoding services currently provide rooftop geocoding. Nevertheless, Rooftop geocoding is becoming widespread because of its improved accuracy over street geocoding. For example, Google claims that 50 million U.S. addresses are available for rooftop geocoding (Google Maps 2008). A comparison between street geocoding and rooftop geocoding is presented in Table 1.

2.2 Online Geocoding Services

With advancements in web technology and location-based services, many online geocoding services are now available to users and applications for geocoding addresses. In fact, many applications, ranging from simple visualization of address locations on a map to complex spatial analysis, require address geocoding with a high level of accuracy. However, no service can exactly match the users and application's requirements or preferences. Some services are open source and available for free where others are commercial. Some well-known and widely-used online geocoding services are Google

Table 1 Comparison between street geocoding and rooftop geocoding

Feature	Street Geocoding	Rooftop Geocoding
Availability	Mostly available for wide range of addresses	Limited availability but expanding rapidly
Mechanism	Interpolation	Direct match with reference database
Match rate	High. Mostly returns result no matter how accurate it is. May provide upper level address	Low. Returns result only if there is a direct match Does not provide any upper level address
Accuracy	Low. Point could be placed in the middle of the road, in different location. May give invalid address	High. Locates the exact address point. Provides valid address
Storage requirement	Low. Data model is compact	High. Needs to cover all addresses in a large geographic extent
Cost of service	Free or low	Comparatively expensive
Usage	High. Most online services use this	Low. Few online services use this
Maintenance	Less effort	Considerable effort

Maps, Bing Maps (previously known as Virtual Earth, VE), Yahoo Maps, MapQuest, and Geocoder.us. A summary of the features of some of these online geocoding services is given in Table 2.

These services differ by the algorithms and reference databases they use and quality of results they produce. Bing Maps offers both street geocoding (VE interpolation or VE_IP) and rooftop geocoding (VE rooftop or VE_RT) services. Google Maps (2008), on the other hand, uses a hybrid of rooftop and street geocoding. Some online services have global coverage, which means they have reference databases of many countries, while others can geocode only in specific geographic areas.

For example, Geocoder.us covers U.S. addresses only (Erle and Walsh 2008) whereas VE_RT rooftop, using rooftop geocoding, can geocode 40% of U.S. addresses and 90% of addresses in Japan (Tetrad 2011). All online geocoding services offer sequential processing, i.e., geocoding one address at a time. Batch processing, where multiple addresses can be submitted, is supported either by commercial versions or through customization (using APIs). Most online geocoding services use TeleAtlas (2009a, 2009b) or NAVTEQ (2009) data as the reference database, some use TIGER/Line files and gazetteers.

The results of geocoding services also differ in terms of match rate and positional accuracy for different address types in different environment settings such as agricultural, commercial, industrial, and residential. Other differences among online geocoding services include maximum number of queries that a service allows per day, metadata attached with the geocoded results, interface types, and alternative matching in case of

Table 2 Features of some widely-used online geocoding services

Service Name	Provider	Service Interface	Reference DB	Coverage	Query Limit/Day	Process Type	Service Type	Meta data	Alternate Match
Virtual Earth[a] (rooftop)	Microsoft	SOAP	NAVTEQ	US-JP	1,000	Single	RT[1]	No	No
Google Maps[b]	Google Inc.	REST-CSV, JSON, or XML	Tele Atlas, NAVTEQ	Global	15,000	Single	RS[2]	Yes	Yes
Geocoder.us[c]	Locator Tech	SOAP; XML-RPC; REST-CSV, RDF	TIGER/Line files	USA	50,000	Single	ST[3]	No	No
MapQuest[d]	AOL	JavaScript API; REST-XML	TIGER/Gazetteers	Global	5,000	Single	ST	No	Yes
Virtual Earth[a] (Interpolation)	Microsoft	SOAP	NAVTEQ	Global	1,000	Single	ST	No	Yes
Yahoo! Maps[e]	Yahoo Inc.	REST-XML	Tele Atlas, NAVTEQ	Global	5,000	Single	ST	Yes	Yes

[1] Rooftop geocoding; [2] Hybrid (rooftop and street geocoding); [3] Street geocoding
[a] *http://www.bing.com/maps*, [b] *http://maps.google.com*, [c] *http://geocoder.us*, [d] *http://www.mapquest.com/*, [e] *http://maps.yahoo.com*

Table 3 Match rate (%) of candidate online geocoding services for different address types

Address Types	Match level	VE_RT	Google	Geocoder .us	MapQuest	VE_IP	Yahoo!
All	Good	76.48	93.64	87.88	80.96	96.32	94.12
(N = 5,000)	Ambiguous	0.00	2.26	5.62	19.04	0.00	5.88
	Unmatched	23.52	4.10	6.50	0.00	3.68	0.00
Agricultural	Good	50.00	80.00	70.00	55.00	90.00	90.00
(N = 20)	Ambiguous	0.00	10.00	25.00	45.00	0.00	10.00
	Unmatched	50.00	10.00	5.00	0.00	10.00	0.00
Commercial	Good	74.34	96.21	89.21	80.17	95.92	91.84
(N = 343)	Ambiguous	0.00	2.62	7.58	19.83	0.00	8.16
	Unmatched	25.66	1.17	3.21	0.00	4.08	0.00
Industrial	Good	65.63	96.88	87.50	65.63	96.88	93.75
(N = 32)	Ambiguous	0.00	0.00	12.50	34.38	0.00	6.25
	Unmatched	34.38	3.13	0.00	0.00	3.13	0.00
Residential	Good	84.51	94.08	90.25	84.02	97.52	95.40
(N = 4,306)	Ambiguous	0.00	1.39	4.13	15.98	0.00	4.60
	Unmatched	15.49	4.53	5.62	0.00	2.48	0.00

mismatched addresses. We conducted the following experiments in our previous works: (1) comparing the geocoding quality of different online geocoding services utilizing street and rooftop geocoding; and (2) evaluating the sensitivity of geocoding quality to different address types (e.g. industrial, residential, commercial, agricultural).

Experiments were conducted using six online geocoding services: VE_RT, Google Maps, VE_IP, Yahoo Maps, MapQuest, and Geocoder.us. Allegheny County, Pennsylvania, which contains about 0.5 million addresses, was used as the study area. Two sets of addresses were sampled. One set contained 5,000 random mixed addresses; another set contained different address type's proportion to the percentage of each address type in the study area. Two metrics, match rates and positional accuracy, were used. Tables 3 and 4 summarize the results of match rate and positional accuracy, respectively. To measure positional accuracy using error distances, an address point baseline, the centroid of the corresponding building footprint or parcel boundary of each sample address, was acquired from the Allegheny County Division of Computer Services GIS Group). Building footprints and parcel boundaries were updated and maintained periodically. According to metadata, both datasets have horizontal accuracy within ±5 feet.

The number of sample addresses in Table 4 is relatively smaller than that in Table 3 because only addresses that could be geocoded by all services were included in the positional accuracy analysis. In Table 3, good match means the service returns a geocoded result at address level while ambiguous match means the service returns a geocoded result at street, city, ZIP code, or country level. As shown in these tables, good match rates tend to have high positional accuracy and ambiguous match rates tend to have lower positional accuracy. In addition, rooftop geocoding services produce higher match rates but lower positional accuracy compared to street geocoding services.

Table 4 Positional accuracy (error distance) of candidate online geocoding services for different address types

Address Types	Error distances Statistics	VE_RT	Google	Geocoder.us	MapQuest	VE_IP	Yahoo!
All	Min (m)	0.1	0.1	0.9	0.9	3.6	0.2
(N = 3,112)	Median (m)	7.7	10.9	54.2	54.9	21.9	34.5
	Max (m)	9,995.6	2,173.0	15,000.9	10,075.6	9,995.6	2,141.2
	95th Percentile	34.6	48.5	272.1	281.9	84.2	168.9
Agricultural	Min (m)	37.6	37.6	76.3	76.2	54.4	78.4
(N = 5)	Median (m)	179.3	214.8	219.2	211.3	195.0	128.8
	Max (m)	227.2	237.7	808.3	610.0	215.5	258.0
	95th Percentile	227.2	237.7	808.3	610.0	215.5	258.0
Commercial	Min (m)	0.9	0.9	6.8	6.7	9.0	1.6
(N = 204)	Median (m)	9.6	10.7	44.9	45.1	24.1	27.9
	Max (m)	698.4	698.4	2,824.3	9,971.6	1,155.0	495.9
	95th Percentile	61.4	72.1	412.1	419.7	100.0	113.7
Industrial	Min (m)	4.5	4.5	14.9	14.9	19.9	13.2
(N = 13)	Median (m)	27.3	27.3	55.6	55.7	39.0	48.9
	Max (m)	124.8	139.8	759.9	759.8	136.9	186.2
	95th Percentile	124.8	139.8	759.9	759.8	136.9	186.2
Residential	Min (m)	0.2	0.2	1.2	1.2	4.1	0.3
(N = 2948)	Median (m)	7.3	9.9	54.3	54.9	21.6	34.5
	Max (m)	10,247.0	8,039.0	32,186.5	15,824.3	10,247.0	1,743.8
	95th Percentile	28.0	43.2	253.4	276.7	69.0	157.7

3 Recommender Algorithm

The recommender is a generalized algorithm that works for given online geocoding services provided that: (1) characteristics of candidate services are tabulated as shown in Table 2; and (2) behaviors of services on match rate and positional accuracy are mined and tabulated. The current version of the algorithm is focused on match rate and positional accuracy as the two key user/application preferences to determine the quality of a service. Match rates are categorized in three classes: good, ambiguous, and unmatched.

Good matches are those geocoded addresses that have a match confidence at address level. Ambiguous matches are those geocoded addresses that have a match confidence either at an upper level like street, city, ZIP code, county, state, or at address level with a warning message. Unmatched is an address that does not return any geocoded result. To rank the services based on match rate, the algorithm considers good match rate first and then considers ambiguous match rate. The higher the match rate, the higher the rank. Positional accuracy is calculated in terms of error distance and is categorized based on descriptive statistics (minimum, median, maximum, 95th percentiles) for each service. The 95th percentiles of the error distance is taken as the ranking criterion to rank the services based on positional accuracy. The lower the error distance, the higher the accuracy. Figure 2 shows the flowchart of the algorithm.

The algorithm utilizes three tables: a Service Summary (SS) table, Match Rate (MR) table, and a Positional Accuracy (PA) table. The SS table, as shown in Table 2, contains relevant characteristics of available online geocoding services that users and/or applications may require. The algorithm uses features like Service Name and Query_Limit (number of queries supported per day) but it could take other features into consideration as well, if necessary. A unique ID field (S_ID) is added to the SS table as the primary key to relate each service with its associated data in the other tables.

The MR and PA tables contain statistics of mining results for match rate and positional accuracy, as shown in Tables 3 and 4, respectively. Inputs to the algorithm are address, address type (agricultural, commercial, industrial, or residential), and the geocoding requirements.

The input can be obtained either interactively through a user interface as shown in Figure 3 or through a file containing the input addresses and requirements. Address type is important because match rate and positional accuracy vary for different types of addresses. In case of unknown address type, the algorithm refers to the statistics of the overall addresses. The user's and/or the application's requirements are prioritized based on match rate and positional accuracy. Users can also specify predefined levels of preferences (High, Medium, Any) for match rate and/or level of accuracy.

Once the inputs are submitted to the recommender, geocoding services that do not satisfy the requirements are filtered out. The filtering process works in three steps. The first step, as shown in Figure 4, filters in those services that pass the query limit by counting the input addresses. Then, the information of the qualified services (*FS_Num*) is retrieved from the SS table.

The second step filters the services based on user's preferences, i.e. match rate and positional accuracy. Three scenarios are handled in this step. First, if the user prefers match rate to accuracy, the algorithm filters the services primarily based on the MR table and then the PA table according to the preferred level of match rate and address type, as shown in Figure 5.

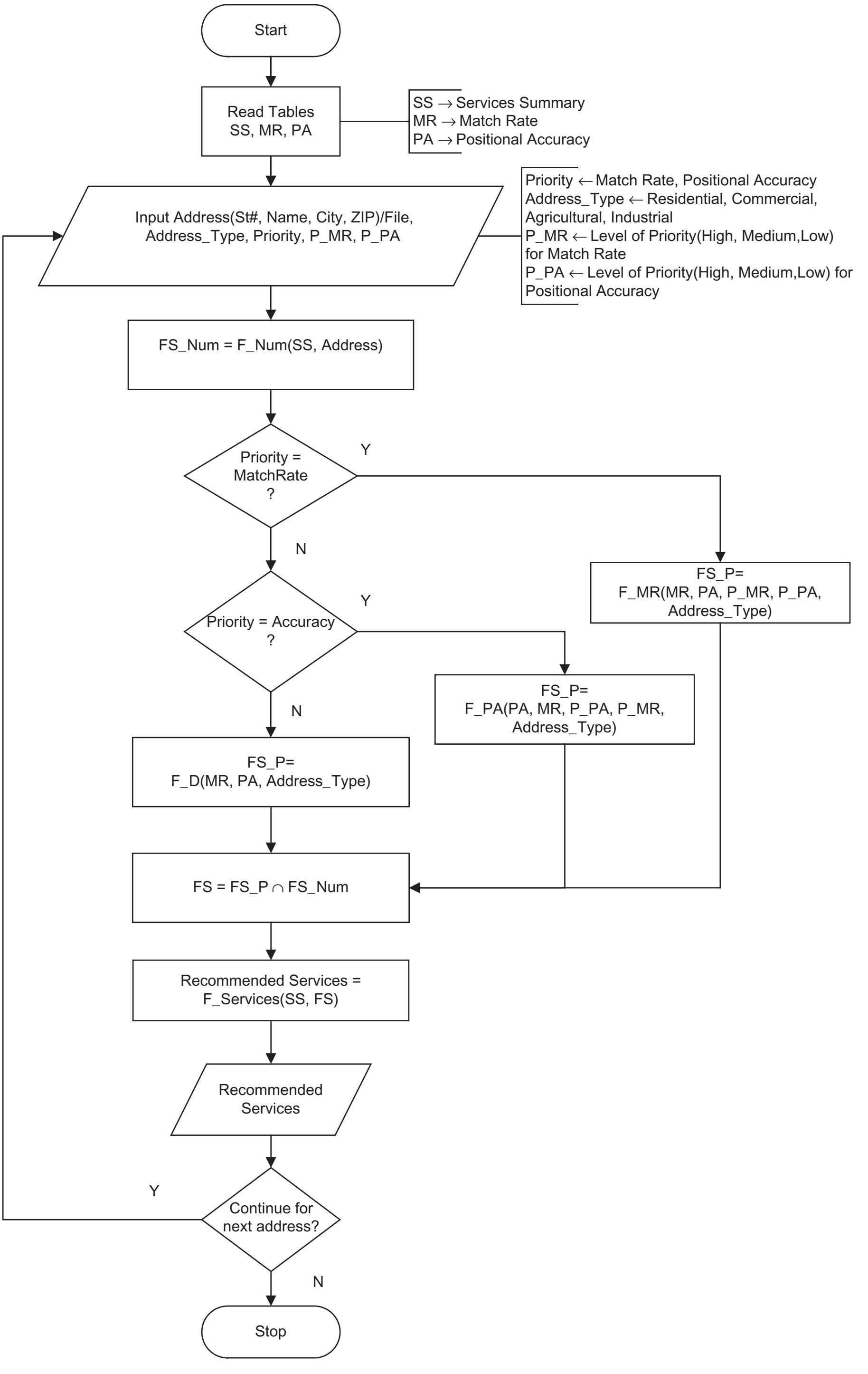

Figure 2 Flowchart of the recommender algorithm

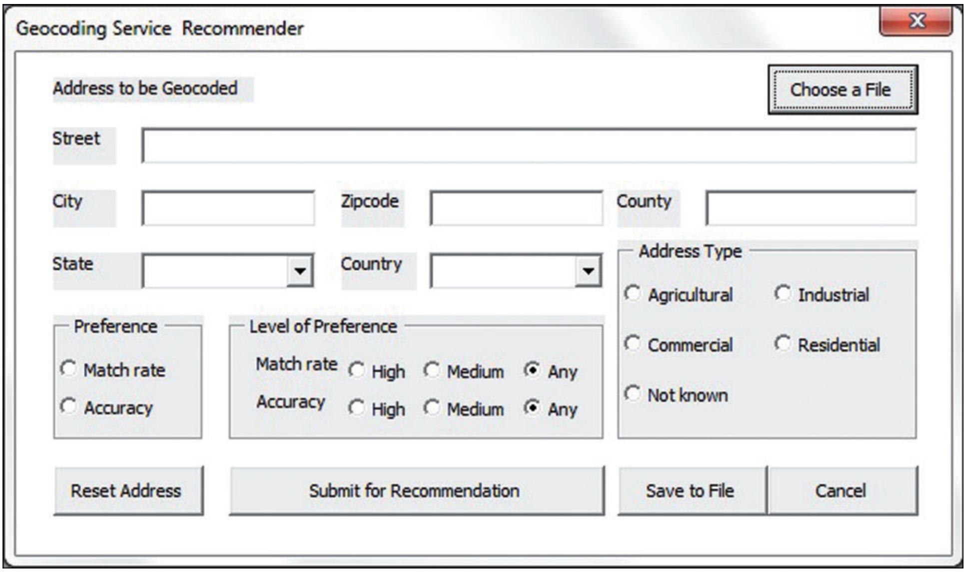

Figure 3 Interface to submit inputs to the Recommender

```
F_Num(SS, add)
    1. C = Count(add)
    2. Res = select S_ID from SS
            where SS.Query_Limit ≥ C
    3. return Res
```

Figure 4 Filtering services based on query limit

```
F_MR(MR, PA, P_MR, P_PA, add_type)
    1. Res1 = Select S_ID from  MR
            where MR.Add_Type = add_type  Λ
            MR.MatchLevel.Good ≥ P_MR
            Order by MR.MatchLevel.Good dsc
    2. Res = Select S_ID from PA
            where PA.S_ID IN RS1 Λ
            PA.Add_Type = add_type  Λ
            PA.95Per ≤ P_PA
            Order by PA.95Per asc
    3. return Res
```

Figure 5 Filtering services with match rate preferred over positional accuracy

The filtering process selects, first from the MR table, services (*Res*1) with a good match rate equal or above the preferred match rate level (*P_MR*) for the specified address type. Then, it further selects services (*Res*) from *Res*1 that have 95th percentile (95Per) of error distances lower or equal to the preferred accuracy level (*P_PA*) for the same address type. Services in *Res* are ranked according to good match rate (descending order) and then the value of 95th percentile of error distances (ascending order).

Second, if the user prefers positional accuracy to match rate, the algorithm filters the services primarily based on PA table and then the MR table according to the preferred

```
F_PA(PA, MR, P_PA, P_MR, add_type)
    1. Res1 = Select S_ID from PA
           where PA.Add_Type = add_type  Λ
           PA.95Per ≤ P_PA
           Order by PA.95Per asc
    2. Res = Select S_ID from MR
           where MR.S_ID IN RS1   Λ
           MR.MatchLevel.Good ≥ P_MR Λ
           MR.Add_Type = add_type  Λ
           Order by MR.MatchLevel.Good dsc
    3. return Res
```

Figure 6 Filtering services with positional accuracy preferred over match rate

```
F_D(add_type)
    1.  Res1 = Select S_ID from  MR
          where MR.Add_Type = add_type  Λ
          Order by MR.MatchLevel.Good dsc
    2.  Res = Select S_ID from PA
          where PA.S_ID IN Res1 Λ
          PA.Add_Type = add_type
          Order by PA.95Per asc
    3.  return Res
```

Figure 7 Filtering services when neither match rate nor positional accuracy is preferred

level of accuracy and address type, as shown in Figure 6. The process selects, first from the PA table, services (*Res*1) with error distance below or equal to the preferred accuracy level (*P_PA*) for the chosen address type. Then from *Res*1, it selects services (*Res*) with a good match rate equal or above the preferred match rate level (*P_MR*) for the same address type. Services in *Res* are ranked according to the value of 95th percentile of error distances (ascending order) and then good match rate (descending order).

In the third scenario, if neither match rate nor positional accuracy is preferred, a default process F_D(), as shown in Figure 7, is followed to filter services. The assumption is that the input address type is known and selected. Services (*Res*1) are selected first from the MR table for the chosen address type. The algorithm then, from the PA table, selects those services (*Res*) in *Res*1 that have the same address type. The services in *Res* are ranked according to the value of good match rate (descending order) and then by the 95th percentile (ascending order) of error distances. In case of unknown address type, a similar procedure as Figure 7 is followed but the address type is selected as "All". The algorithm selects services based on match rate and positional accuracy over all addresses. Step 2 finally returns the selected services (*Res*) to FS_P which are passed on to the last step.

In the final step, services (*FS*) that satisfy all the requirements are found by intersection of *FS_P* and *FS_Num*, which were determined in earlier steps. The details of each service in *FS* for recommendation are extracted from SS table, as shown in Figure 8. The recommended services are presented to the user in rank order.

4 Simulation Results and Discussion

To demonstrate the outcome of the recommendation algorithm, we simulated geocoding various addresses belonging to different environment settings and possible preferences. In

```
F_Services(SS, List)
    Services = [ ]
    For each id IN List
        name = Select ServiceName from SS
                where SS.S_ID = id
        Services.append(name)
    return Services
```

Figure 8 Extracting recommended services

this simulation, all possible inputs (e.g. address type, preferences) were considered. As the current version of the Recommender algorithm depends on the type of address (agricultural, commercial, industrial, and residential), the user's and/or the application's preferences, and the behavior of each online geocoding service, the actual addresses were not considered as input for simulation. In the simulation, six widely-used online geocoding services were investigated. The algorithm recommends services based on two metrics (preferences), match rate and positional accuracy. Match rate was calculated as the percent ratio of the number of geocoded addresses to the total number of addresses submitted. The positional accuracy was taken as a measure (in meters) of the closeness of each geocoded point to the "true" location. For each metric, three levels of measurement (High, Medium, Other) are defined. For match rate, the levels of measurement are defined as High $\geq$ 90%, 80% $\leq$ Medium < 90%, and Other < 80%. For accuracy, the levels (error distance in meters) of measurement are defined as High $\leq$ 50, 50 > Medium $\geq$ 100, and Other > 100. In the simulation, we coded address types as follows: agricultural = 1, commercial = 2, industrial = 3, residential = 4, and any = 0.

The simulation was developed using Python and SQLite 3 was used to handle the reference tables. The computational complexity of the simulation is not high because, in this simulation, six geocoding services were compared by using their result statistics for five different address types. The current simulation provides three options (metrics) as preferences of geocoding quality: match rate, accuracy, or any, where each metric (option) has three levels of measurement; High, Medium, and Other. With these, there are $6 \times 5 \times 3 \times 3 = 270$ different input possibilities. Fifteen out of 270 possible inputs were randomly selected to show how the algorithm recommends different online services depending on different environment settings and preferences.

4.1 Data and Method

Six well-known, widely-used and freely available online geocoding services were employed in the simulation: VE_RT, Google Maps, Geocoder.us, MapQuest, VE_IP, and Yahoo Maps. The results of mining services in our previous works, which included statistics on match rate and accuracy based on address points, building footprints, parcel boundaries, and street centerlines in Allegheny County, Pennsylvania, were used as a baseline. These statistics were based on a total of 5,000 randomly selected addresses in Allegheny County, Pennsylvania.

The simulation is run for all input addresses. Fifteen sample input addresses with different address types and preferences (as shown in Table 5) were randomly chosen for

Table 5 Sample input addresses and preferences

Address	AddressType	Priority	P_MR	P_PA
A1	1	MR	H	H
A2	2	MR	H	M
A3	3	MR	M	M
A4	4	MR	H	M
A5	0	MR	M	M
A6	1	PA	H	H
A7	2	PA	M	H
A8	3	PA	M	M
A9	4	PA	M	H
A10	0	PA	M	M
A11	1	NA	X	X
A12	2	NA	X	X
A13	3	NA	X	X
A14	4	NA	X	X
A15	0	NA	X	X

demonstration. Since in the current version of the simulation addresses can be in any format, the selected addresses are coded as A1 to A15. The Priority column indicates the user's and/or the application's preference: MR means match rate is preferred; PA means positional accuracy is preferred; and NA means no preference is specified. The P_MR and P_PA columns indicate the level of preference for match rate and positional accuracy, respectively. H, M, and X represent High, Medium, and Other level of preference, respectively.

4.2 Results and Discussion

The recommendation results for the sample addresses, along with their input parameters, are summarized in Table 6. Columns R1 through R6 are recommended services ranked from high to low. Value X under recommended services means that no service is available for the given inputs and preferences. The recommended online geocoding services are ranked according to the preferences and priorities of user/application for each given input address. For example, for address A6, address type is 1 (agricultural), priority is PA (positional accuracy), P_PA (preferred accuracy level), and P_PR (preferred match rate level) are H (High), no services are found. A different example is address A15 for which address type is 0 (Any), priority is NA (any), P_PA (preferred accuracy level), and P_PR (preferred match rate level) are X (any), all services are recommended in rank order (high to low) as VE_IP, Yahoo, Google, Geocoder.us, MapQuest, and VE_RT.

In order to show that the algorithm can recommend different services according to various preferences, the preferences of some inputs (as shown in Table 7) are modified. The recommendation results for the modified inputs are shown in Table 8.

The overall result also shows that when match rate is preferred, Geocoder.us and MapQuest are less likely to be recommended or are recommended with a lower rank

Table 6 Online geocoding services recommended for addresses of different types and preferences

Address	Preferences				Recommended Services (Ranked)					
	Address Type	Priority	P_MR	P_PA	R1	R2	R3	R4	R5	R6
A1	1	MR	H	H	X					
A2	2	MR	H	M	Google	VE_IP				
A3	3	MR	M	M	X					
A4	4	MR	H	M	VE_IP	Google				
A5	0	MR	M	M	VE_IP	Google				
A6	1	PA	H	H	X					
A7	2	PA	M	H	X					
A8	3	PA	M	M	X					
A9	4	PA	M	H	VE_RT	Google				
A10	0	PA	M	M	VE_RT	Google	VE_IP			
A11	1	NA	X	X	VE_IP	Yahoo	Google	Geocoder.us	MapQuest	VE_RT
A12	2	NA	X	X	Google	VE_IP	Yahoo	Geocoder.us	MapQuest	VE_RT
A13	3	NA	X	X	VE_IP	Google	Yahoo	Geocoder.us	MapQuest	VE_RT
A14	4	NA	X	X	VE_IP	Yahoo	Google	Geocoder.us	VE_RT	MapQuest
A15	0	NA	X	X	VE_IP	Yahoo	Google	Geocoder.us	MapQuest	VE_RT

Table 7 Sample input addresses with modified preferences

Address	AddressType	Priority	P_MR	P_PA
A1	1	MR	H	H
A2	2	MR	H	M
A3	3	PA	M	M
A4	4	MR	H	M
A5	0	MR	M	M
A6	1	PA	H	H
A7	2	MR	H	H
A8	3	PA	M	M
A9	4	PA	M	H
A10	0	PA	H	H
A11	1	NA	X	X
A12	2	NA	X	X
A13	3	PA	H	H
A14	4	NA	X	X
A15	0	MR	H	X

than others. On the other hand, VE_IP and Google are recommended with a higher rank than others. For positional accuracy preference, VE_RT and Google are recommended with a higher rank than others whereas Geocoder.us and MapQuest performed poorly. The reason behind this poor performance by Geocoder.us and MapQuest may be that both use TIGER/Line as reference database while most other services use commercial reference databases which are comparatively more accurate.

5 Summary and Future Research

Selecting a suitable geocoding service that meets specific requirements of a user and/or application is a challenging task. This is because, on the one hand, the user's and/or application's requirements and preferences are heterogenous, while on the other hand, numerous online geocoding services with different capabilities, algorithms, and reference databases are available. Such differences are the main reasons for obtaining different geocoded results for the same address by different online geocoding services. In this article, we presented a recommender algorithm that can recommend optimal online geocoding services to meet the user's and/or application's requirements. The recommendation is based on characteristics and metrics (positional accuracy and match rate) of available online geocoding services. A simulation was performed to demonstrate the recommendation results of the algorithm. Simulation result validated the rationality of the recommendations.

In future work, input addresses will automatically be parsed so that the recommender can decide the type of addresses, i.e., users will not be required to know the address type. Also, given that online geocoding services are frequently updated and improved, a new technique that automatically detects changes in the behavior of each online service will be developed.

Table 8 Recommendations for same address set with different preferences

	Preferences				Recommended Services (Ranked)					
Address	Address Type	Priority	P_MR	P_PA	R1	R2	R3	R4	R5	R6
A1	1	MR	H	H	X					
A2	2	MR	H	M	Google	VE_IP				
A3	3	PA	M	H	X					
A4	4	MR	H	M	VE_IP	Google				
A5	0	MR	M	M	VE_IP	Google				
A6	1	PA	H	H	X					
A7	2	MR	M	H	X					
A8	3	PA	M	M	X					
A9	4	PA	M	H	VE_RT	Google				
A10	0	PA	H	H	Google					
A11	1	NA	X	X	VE_IP	Yahoo	Google	Geocoder.us	MapQuest	VE_RT
A12	2	NA	X	X	Google	VE_IP	Yahoo	Geocoder.us	MapQuest	VE_RT
A13	3	MR	H	H	X					
A14	4	NA	X	X	VE_IP	Yahoo	Google	Geocoder.us	VE_RT	MapQuest
A15	0	MR	H	X	VE_IP	Yahoo	Google			

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